1.1 Managing relationships through digital identity

Ever since the first timeshare systems were invented, developers and systems engineers have been looking for ways to restrict access to various components of computer software systems for different people. The move to the web and then cloud-based applications magnified that problem as more and more people, organizations, and services have gone online. Getting resources and data into the right hands is a significant strategic advantage, as is keeping them out of the wrong ones.

Even so, for many years, Identity and Access Management (IAM) has been more focused on the "I" than the "AM." But advances in access control based on policy and the emphasis on "zero trust" in security have brought access management to the fore. Successful access management requires a holistic approach to digital identity, carefully considering the goals and needs of the systems it will serve.

As the Target breach points out, digital identity is foundational not just in securing computer systems but also in enabling features that make them usable. People often think of things like user IDs, passwords, accounts, and so on when the subject of digital identity comes up. But let's start with why we build digital identity systems: relationships.

Every digital identity system is built to *manage relationships* of some kind. The system is controlled by some entity that uses it to manage the relationships they care about. Managing relationships requires that the controller *recognize*, *remember*, and *relate to* other entities. (I'll often use the word *entity* in this book to refer generically to subjects or principles that could be a person, an organization, a service, a thing, or an AI agent.) In the physical world, we effortlessly recognize, remember, and relate to everyone and everything around us. As human beings, we have an identity system embedded deep within us that allows us to create relationships with the world around us. Online, it's not so easy.

One of the fundamental features of the online world is that we interact with others at a distance. This is called the *proximity problem* and is the first problem any online system must solve. That's why digital identity is a core